BSc (Hons) in Data Science
IT3101 - DWBI - Assignment
Design and Implementation of a Data Warehouse and Business Intelligence Solution

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Task 1: Dataset Selection and Business Scenario Identification

1.1 Business Domain and Scenario

Domain (E-commerce Logistics & Supply Chain), OLTP system description, ~99,441 orders spanning September 2016 to October 2018, problem addressed (delivery lead time optimization, freight cost management, geographical fulfillment analysis).

1.2 Dataset Overview

Dataset Source: Kaggle (Olist Brazilian E-commerce Dataset - Brazilian E-Commerce Public Dataset by Olist)
Dataset Format: Detailed attributes of the 9 CSV files (olist_customers_dataset.csv, olist_orders_dataset.csv, olist_order_items_dataset.csv, olist_products_dataset.csv, olist_sellers_dataset.csv, olist_order_payments_dataset.csv, olist_order_reviews_dataset.csv, olist_geolocation_dataset.csv, product_category_name_translation.csv), relationships, and data preparation steps (missing value imputation, type casting, duplicate removal).

The Olist Brazilian E-commerce dataset is highly suitable for an OLTP to OLAP DWBI solution because its structural complexity and transactional volume perfectly mirror a real-world relational database environment requiring analytical optimization. It fulfills all the criteria for an end-to-end Data Warehouse project through the following key characteristics:

Highly Normalized Operational Schema (OLTP) In its raw form, the dataset is fragmented across nine distinct, highly normalized CSV files (mimicking a Third Normal Form relational database). Because customer data, product specifications, order timestamps, and transactional items are stored separately, running direct analytical queries (such as calculating the average freight cost by product category per state) would require highly inefficient, resource-heavy multi-table joins. This structural bottleneck justifies the need for an OLAP Data Warehouse to denormalize and streamline the data.

Clear Dimensional Mapping The attributes natively support a standard Star or Snowflake Schema. The data can be seamlessly categorized into classical dimension tables:

- **Time/Date Dimension:** Extracted from the olist_orders_dataset.csv timestamps (purchase, approval, and delivery dates).
- **Geography Dimension:** Derived from the olist_geolocation_dataset.csv and customer/seller zip codes, enabling spatial analysis at the state and city levels.
- **Product & Entity Dimensions:** Sourced from the customer, seller, and product datasets to provide descriptive context to the sales transactions.

Granular Fact Table Potential With over 112,000 recorded order items, the dataset provides a highly granular base for a Fact Table. This allows for the calculation of robust, additive, and semi-additive business measures, including total item revenue, freight value, delivery delay duration, and customer review scores at the individual item grain.

Real-World ETL Complexity The data presents realistic inconsistencies that perfectly demonstrate the necessity of the Extraction, Transformation, and Loading (ETL) phase. It requires handling missing delivery dates for canceled orders, mapping Portuguese product categories to English using the translation table, standardizing geographical text fields, and calculating derived metrics (such as SLA breaches by subtracting estimated delivery dates from actual delivery dates).

High Business Intelligence Value The dataset directly addresses tangible business problems within the logistics and supply chain domain. By migrating this data into an OLAP structure, it enables the creation of powerful Business Intelligence dashboards to track KPI trends over a two-year period, identify regional shipping bottlenecks, and analyze how delivery delays impact overall customer satisfaction.

Task 2: Data Source Identification and Preparation

2.1 Source Descriptions

The solution utilizes multiple external source files representing a highly normalized operational e-commerce and logistics environment. All data is extracted from flat **CSV** files, which emulate tables in a relational database management system.

2.2 Available Attributes

The following core datasets and their specific attributes are used to build the analytical solution:

- **olist_orders_dataset.csv:** Acts as the central transactional hub tracking order lifecycles.
    - *Attributes:* order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date.

- **olist_order_items_dataset.csv:** Contains granular line-item details for shipping and pricing.
    - *Attributes:* order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value.

- **olist_customers_dataset.csv:** Stores demographic and location routing data for the buyers.
    - *Attributes:* customer_id, customer_unique_id, customer_zip_code_prefix, customer_city, customer_state.
- **olist_sellers_dataset.csv:** Provides the origin location details for fulfillment.
    - *Attributes:* seller_id, seller_zip_code_prefix, seller_city, seller_state.
- **olist_products_dataset.csv:** Contains physical product dimensions necessary for logistics.
    - *Attributes:* product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm.
- **product_category_name_translation.csv:** A reference table used for data standardization.
    - *Attributes:* product_category_name, product_category_name_english.

2.3 Relationships Between Sources

To form a cohesive data model, these fragmented CSV files share specific primary and foreign key relationships that must be joined during the integration phase:

- The olist_orders_dataset.csv links to olist_customers_dataset.csv via the **customer_id** key.

- The olist_orders_dataset.csv links to olist_order_items_dataset.csv via the **order_id** key, expanding the data from an order grain to an item grain.

- The olist_order_items_dataset.csv bridges to both the olist_sellers_dataset.csv (via **seller_id**) and the olist_products_dataset.csv (via **product_id**).

- The olist_products_dataset.csv links to the translation file via **product_category_name** to resolve language inconsistencies.

2.4 Data Preparation Steps

Before loading the data into the analytical warehouse, several preparation and transformation steps must be executed to ensure data quality:

- **Missing Value Handling:** Incomplete delivery lifecycles (e.g., canceled orders missing an order_delivered_customer_date) are filtered out or imputed with default values so they do not skew average delivery time calculations. Missing product categories are labeled as "Unknown".

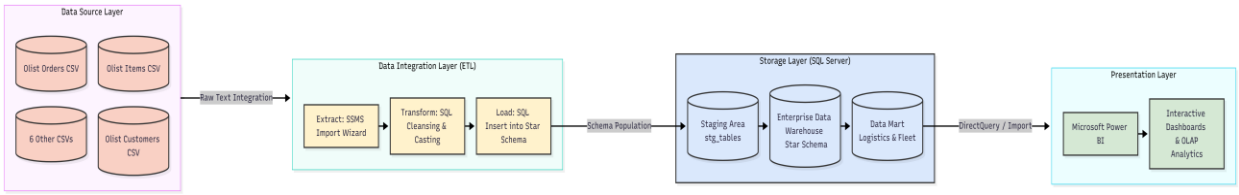
- **Data Type Conversion:** Transactional timestamps are stored as text strings in the CSV files. These must be explicitly cast to datetime objects to allow for time-based aggregation and duration calculations (e.g., subtracting purchase date from delivery date).
- **Data Cleansing & Standardization:** Geographic attributes, such as city names, are cleansed of trailing whitespace, stripped of special characters, and standardized to uppercase to prevent duplicate grouping in OLAP reporting.
- **Data Enrichment:** The Portuguese product names are merged with the translation CSV to create a standardized English category column, improving the readability of the final BI dashboards.

2.5 Integration into a Unified Analytical Environment

Integrating these diverse operational files into a unified analytical environment is achieved by utilizing an ETL (Extract, Transform, Load) pipeline. A Python-based script using the Pandas library reads each CSV into memory as a separate dataframe.

Once the data type conversions and cleansing rules are applied, the dataframes are horizontally integrated using inner and left joins based on their defined relationships. This staging process denormalizes the scattered OLTP data into a unified, structured format (a Star Schema). The resulting cleansed dimensional tables and central fact table are then loaded into an Enterprise Data Warehouse (such as Microsoft SQL Server or MySQL), providing a single source of truth optimized for high-speed querying by business intelligence tools like Power BI.

Task 3: Data Warehouse Architecture Design



3.1 Data Source Layer

The Data Source Layer represents the origin of the operational data. For this Business Intelligence solution, the primary sources are external flat files representing the Olist e-commerce operational transaction processing (OLTP) system. This includes nine highly normalized CSV files capturing distinct business entities: orders, granular line items, customer demographics, seller locations, product specifications, geolocation coordinates, payment records, customer reviews, and category translations.

3.2 Data Integration Layer

The Data Integration Layer governs the Extract, Transform, and Load (ETL) processes required to move data from the operational state to the analytical state.

- **Extraction:** Data is systematically extracted from the external CSV files and ingested into Microsoft SQL Server Management Studio (SSMS) using bulk import operations, bypassing strict data typing to ensure zero data loss.
- **Transformation:** SQL-based data manipulation handles the core business logic. This includes casting unstructured string data (NVARCHAR(MAX)) into strict analytical data types (DECIMAL, DATETIME), imputing missing delivery timestamps, trimming text spaces, standardizing uppercase geography fields, and joining translation mapping tables.
- **Loading:** The cleansed, denormalized data is systematically inserted into the structured dimension and fact tables, maintaining strict primary and foreign key constraints for referential integrity.

3.3 Storage Layer

The Storage Layer is the centralized repository for the analytical data, hosted within a relational database management system (Microsoft SQL Server) and divided into three logical zones:

- **Staging Area:** A temporary holding zone containing stg_ prefix tables. It stores the exact replica of the raw CSV files without constraints, acting as a buffer to prevent operational load failures.
- **Enterprise Data Warehouse:** The core analytical repository designed using dimensional modeling principles. It utilizes a Star Schema architecture, organizing the transformed data into surrounding dimension tables (e.g., dim_customer, dim_product, dim_date) and a central, highly granular fact table (fact_order_items) to track measurable logistics events.
- **Data Mart:** A department-specific subset derived from the **Enterprise Data Warehouse**. For this solution, a "Logistics & Fleet Operations Data Mart" is provisioned to isolate freight costs, delivery delays, and spatial routing metrics specifically for Supply Chain users.

3.4 Presentation Layer

The Presentation Layer acts as the user interface for Business Intelligence consumption. Microsoft Power BI is connected directly to the SQL Server Data Mart via Direct Query or Import mode. This layer provides OLAP capabilities such as filtering, slicing, drilling down from yearly to daily grains, and rolling up regional data and hosts the interactive BI dashboards used by executives to extract actionable logistics insights.

Task 4: Dimensional Data Warehouse Design and Implementation



4.1 Fact Table Definition

- Business Process: E-commerce order fulfillment and logistics tracking.
- Grain: One row per individual product item shipped within a specific order.
- Measures: The quantitative metrics being analyzed: price, freight_value, and delivery_delay_days.
- Foreign Keys: The surrogate keys linking the fact table to the dimensions: customer_key, seller_key, product_key, and order_purchase_date_key.

4.2 Students should implement

- Star Schema or Snowflake Schema - A **Star Schema** was implemented rather than a Snowflake schema. The central fact_order_items table connects directly to denormalized dimension tables (dim_customer, dim_seller, dim_product, dim_date). This design avoids complex, deep joins, ensuring fast query performance when aggregating logistics data in the Presentation Layer.

- **Primary Keys (PK):** Integer-based surrogate keys (e.g., customer_key, seller_key) were created using the IDENTITY property in SQL Server for each dimension table.

- **Foreign Keys (FK):** These surrogate keys are stored in the `fact_order_items` table and explicitly linked back to the dimension tables using formal `FOREIGN KEY` constraints to enforce strict referential integrity.

- **Relationships and Hierarchies:**
    - **Relationships:** The architecture enforces **one-to-many (1:N)** relationships. A single record in a dimension table (like one specific product) links to many transactional records in the central fact table.

    - **Hierarchies:** Natural drill-down paths are built directly into the denormalized dimension tables to enable BI slicing. For example, dim_customer and dim_seller contain a geographic hierarchy of **State -> City -> Zip Code**, while dim_date contains a temporal hierarchy of **Year -> Quarter -> Month -> Date**.

- Table Descriptions:
    - **dim_customer:** A dimension table containing customer geographic hierarchies (State, City, Zip Code) used to analyze buyer locations and regional demand.
    - **dim_seller:** A dimension table containing seller geographic locations, enabling the analysis of origin points for freight and shipping routes.
    - **dim_product:** A dimension table storing physical product attributes (weight, height, length) and English-translated categories to evaluate how product types impact logistics and shipping costs.
    - **dim_date:** A temporal dimension table generated from purchase timestamps, allowing the BI tool to filter and trend logistics performance by Year, Quarter, Month, and Day of the Week.
    - **fact_order_items:** The central transactional table storing the quantitative measures for each shipped item (price, freight_value, delivery_delay_days) alongside the foreign keys that link back to the dimension tables.

- Design Assumptions:
    - **Performance Optimization:** A Star Schema design was chosen over a Snowflake design. By denormalizing location data directly into the Customer and Seller dimensions, we reduce the number of required joins, significantly improving dashboard rendering speeds in Power BI.

    - **Data Scope (Completed Cycles Only):** It is assumed that canceled or unavailable orders skew logistics performance metrics. Therefore, only records with an order_status of 'delivered' are loaded into the fact table.

    - **System Independence:** It is assumed the operational system's natural keys might change or contain anomalies. Therefore, integer-based surrogate keys (_key) are used exclusively for primary and foreign key relationships in the data warehouse to guarantee referential stability.

Task 5: ETL Process Development

5.1 ETL Workflow

- The ETL pipeline extracts raw operational data from the Olist CSV files into a SQL Server Staging Area. Data is initially ingested as raw strings to prevent import errors caused by dirty data. The data is then cleaned, integrated, and transformed in-database using T-SQL scripts before being loaded into the optimized Star Schema tables for Business Intelligence consumption.

5.2 Data Extraction

- The Extraction phase was designed to prioritize data preservation and system stability.
    - Data was extracted from the nine raw operational CSV files representing the Olist e-commerce system.
    - The data was loaded into a SQL Server Staging Area (Olist_DWBI_Staging) without enforcing strict data types or primary keys (utilizing NVARCHAR(MAX) for text and coordinates). This "all-string staging" approach prevented import failures caused by messy operational data or stray characters.

5.3 Data Transformation

- Once in the staging area, T-SQL (Transact-SQL) scripts were utilized to clean, standardize, and enrich the data for analytical processing. The following key transformations were applied:

    - **Data Type Casting:** Raw string data was safely converted into strict analytical formats using the TRY_CAST() function. For example, product weights and freight values were cast to DECIMAL(10,2), and purchase timestamps were cast to DATE.

    - **Text Standardization:** Inconsistent text entries in the customer and seller location fields were standardized using UPPER() and TRIM() functions to ensure accurate grouping in the BI dashboards.

- **Data Integration (Translation):** A lookup mapping transformation was performed using a LEFT JOIN. The original Portuguese product categories (e.g., *beleza_saude*) were successfully translated into English (*health_beauty*) using the stg_product_category_name_translation table.

- **Handling Missing Values:** The COALESCE() function was applied during the product translation phase to replace missing or unmapped categories with the string 'UNKNOWN', preventing NULL values from breaking dashboard visuals.

- **Derived Metrics (Business Logic):** A new performance metric, delivery_delay_days, was dynamically calculated using the DATEDIFF() function to find the variance between the estimated delivery date and the actual delivery date.

5.4 Data Loading

- The transformed data was loaded into the final Enterprise Data Warehouse Star Schema.
  - **Dimension Loading:** Data was inserted into dim_customer, dim_seller, dim_product, and dim_date using SELECT DISTINCT to eliminate duplicate records. SQL Server's IDENTITY(1,1) property automatically generated sequential Surrogate Keys (_key) for each row.

  - **Fact Loading:** The fact_order_items table was loaded by performing INNER JOIN and LEFT JOIN operations between the staging transaction tables and the newly created dimension tables. This process successfully swapped the operational alphanumeric business keys for the optimized integer-based surrogate keys.

5.5 Validation Results

- Post-load validation was performed using COUNT(*) aggregation queries to ensure the fact table row counts aligned with the expected volume of completed ('delivered') orders. Validation queries (SELECT TOP 100) confirmed that the surrogate keys auto-incremented properly, translations mapped successfully, and no orphan records existed in the central fact table.

Task 6: Data Mart Development

6.1 Purpose of the Data Mart

The purpose of Logistics Data Mart is to provide a focused, pre-aggregated subset of the Enterprise Data Warehouse tailored specifically for tracking order fulfillment and shipping efficiency. By isolating relevant logistics metrics such as delivery delays and freight costs from broader retail sales data, it creates a streamlined, single-subject repository designed exclusively to monitor supply chain performance.

6.2 Target Users

The primary users are Supply Chain Managers, Regional Logistics Coordinators, and Fleet Operations personnel. These stakeholders require targeted, day-to-day insights into delivery workflows, shipping expenditures, and geographic performance bottlenecks without being overwhelmed by unrelated marketing or product catalog data.

6.3 Analytical benefits

- **Enhanced Query Performance:** By pre-aggregating complex calculations (like the sum of freight costs and the average of delivery delays) at the database level, the Data Mart drastically reduces the computational load on Business Intelligence tools. Power BI can query a small summary view instead of scanning over 100,000 raw fact rows, resulting in near-instant dashboard rendering.

- **Simplified User Experience:** The Data Mart masks the structural complexity of the underlying Star Schema. Non-technical business users can drag and drop fields from a single, unified view to build their reports, removing the need for them to understand complex SQL joins or primary, foreign key relationships.

- **Targeted Decision-Making:** It delivers immediate, actionable intelligence aligned with specific operational goals. Users can instantly analyze historical trends, such as identifying which customer states consistently experience the highest delivery delays month-over-month, allowing for rapid resource reallocation.

```sql
create view logistic_data_mart as
select
    d.year,
    d.month,
    c.customer_state,
    COUNT(distinct f.order_id) as total_orders,
    SUM(f.freight_value) as total_freight_cost,
    AVG(CAST(f.delivery_delay_days AS DECIMAL(10,2))) AS avg_delivery_delay_days
FROM fact_order_items f
INNER JOIN dim_customer c
    ON f.customer_key = c.customer_key
INNER JOIN dim_date d
    ON f.order_purchase_date_key = d.date_key
GROUP BY
    d.year,
    d.month,
    c.customer_state;

SELECT TOP 15 * FROM logistic_data_mart ORDER BY total_orders DESC;
```

100 % 6 0

Results Messages

	year	month	customer_state	total_orders	total_freight_cost	avg_delivery_delay_days
1	2018	8	SP	3164	56101.49	-5.576912
2	2018	5	SP	3138	53873.37	-10.786029
3	2018	4	SP	3002	52178.35	-12.471903
4	2018	1	SP	2975	49681.06	-12.504291
5	2018	3	SP	2971	52077.17	-7.868328
6	2017	11	SP	2899	50210.69	-8.829850
7	2018	6	SP	2738	51462.75	-16.512507
8	2018	7	SP	2715	50173.57	-9.125806
9	2018	2	SP	2632	43121.21	-9.891311
10	2017	12	SP	2295	38150.77	-14.092378
11	2017	10	SP	1723	30096.72	-11.196165
12	2017	8	SP	1665	27346.76	-12.214509
13	2017	9	SP	1576	27315.66	-10.683736
14	2017	7	SP	1543	26310.86	-11.531723
15	2017	5	SP	1363	23449.88	-11.299674

Figure 1: Execution and validation of the Logistics Data Mart (logistic_data_mart) showing pre-aggregated regional delivery metrics.

Task 7: OLAP Analysis and Business Intelligence Dashboard Development

7.1 Report Page 1: Executive Summary

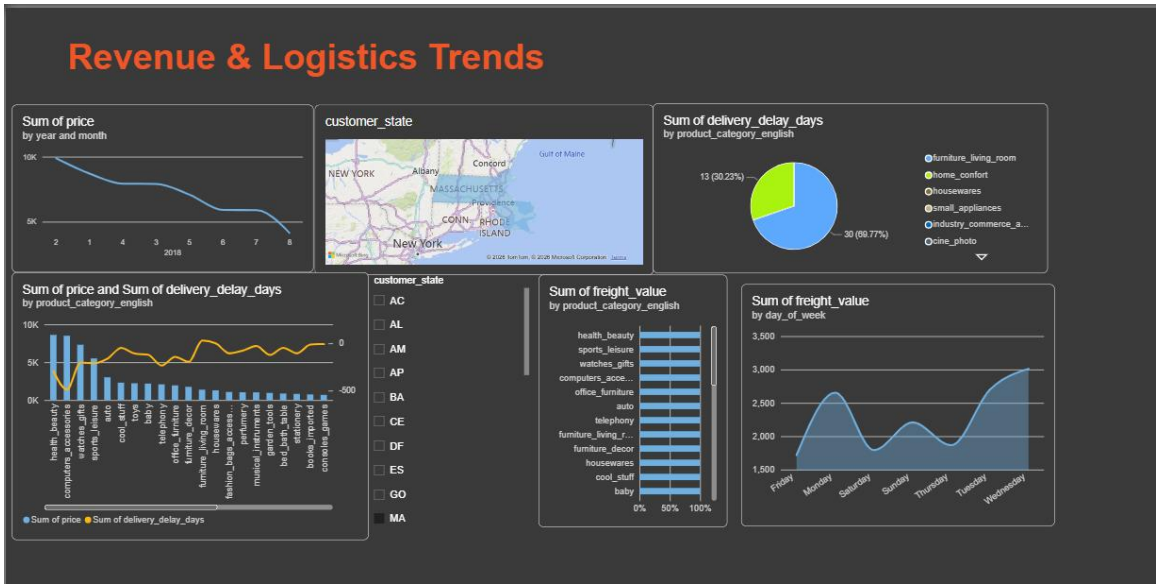


- **Objective:** This page fulfills the assignment requirement to establish high-level KPIs, summary cards, and overall performance indicators.

- **Design & Features:** The dashboard displays five summary cards tracking Total Revenue (13.22M), Average Freight Value (19.95), Total Orders (132K), Average Delivery Delay (-15.21 days), and Total Freight Cost (2.20M).

- **Business Insight:** A horizontal bar chart breaks down total revenue by product category, highlighting "health_beauty" and "watches_gifts" as the top-performing categories. A donut chart visualizes revenue distribution across customer states, showing the "SP" state generating the largest share at 38.33% (5.07M). A timeline chart tracks the sum of price and order items across purchase dates.

7.2 Report Page 2: Trend Analysis

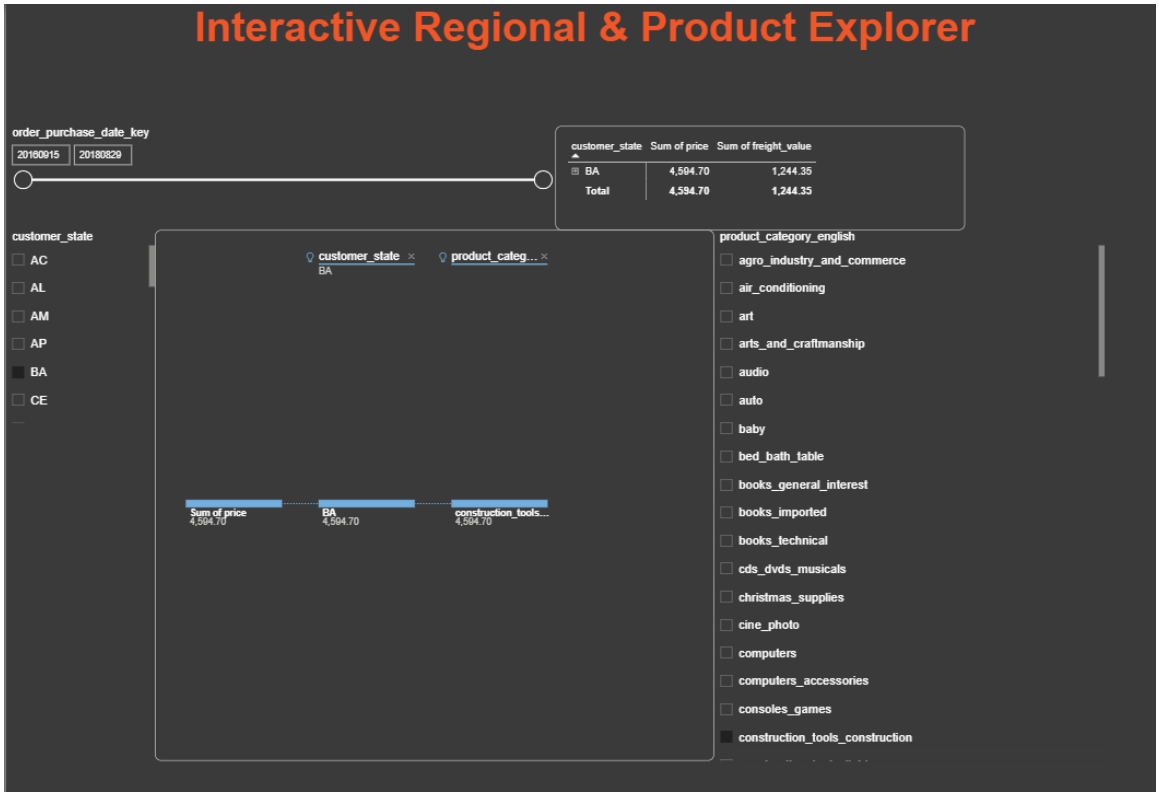


- **Objective:** This page fulfills the requirement for time-based analysis and comparison charts.

- **Time-Based Analysis:** A line chart visualizes the sum of price by year and month, illustrating a downward revenue trend through 2018. A separate line chart tracks the sum of freight value across different days of the week.

- **Comparison Charts:** A clustered column and line chart compares total revenue (columns) against total delivery delay days (line) for each product category. The page also includes a geographical map of customer states, a donut chart analyzing delivery delays by category, and a 100% stacked bar chart analyzing freight value by category. A slicer panel allows users to filter these trends by specific customer states.

7.3 Report Page 3: Interactive Analysis



- **Objective:** This page demonstrates interactive data analysis through filters, slicers, drill-downs, and roll-ups.

- **Interactive Filters & Slicers:** The left panel features a date range slicer spanning from September 15, 2016, to August 29, 2018. It also includes individual checkbox slicers for both customer_state and product_category_english.

- **Roll-up & Drill-down:** A matrix visual displays aggregated metrics for selected regions, such as showing a sum of price of 4,594.70 and a sum of freight value of 1,244.35 for the state of "BA".

- **Interactive Analysis:** A Decomposition Tree interactively breaks down the total sum of price (4,594.70) by the specific selected customer_state (BA), allowing users to click and expand the data branches dynamically.

Task 8: Business Insights and Recommendations

Based on the dashboard outputs, the following insights and recommendations are presented:

1. **Regional Comparison: Market Concentration in SP and RJ**
    - **Observation:** The regional revenue donut chart reveals a significant market concentration, with the 'SP' (São Paulo) state generating 38.33% (5.07M) of total revenue, followed by 'RJ' (Rio de Janeiro) at 13.31%. Combined, these two states account for over half of all business income.

    - **Business Decision-Making:** This insight directs resource allocation. The company should strategically locate their primary fulfillment centers near SP and RJ to reduce shipping distances and costs. Marketing teams should allocate the largest portion of their budget to these proven, high-converting regions to maximize Return on Ad Spend (ROAS).

2. **Sales Trends: Dominance of Health & Beauty**
    - **Observation:** The categorical bar chart indicates that health_beauty and watches_gifts are the most profitable product categories, outperforming historically strong e-commerce sectors like electronics and computers.

    - **Business Decision-Making:** Procurement and inventory management teams must prioritize stock levels for these top-tier categories to prevent stockouts. Sales teams can leverage this data to negotiate better wholesale rates with suppliers in the health and beauty sector, or create high-margin bundled promotions (e.g., combining a watch with a fragrance) to increase the average order value.

3. **Operational Performance: Over-performing Logistics**
    - **Observation:** The overall average delivery delay is -15.21 days. A negative number in logistics signifies that packages are consistently arriving well ahead of the estimated delivery date shown to the customer.

    - **Business Decision-Making:** While early delivery is positive, a 15-day discrepancy suggests the business is heavily over-estimating shipping times at checkout. Management should tighten these estimated delivery windows. Displaying faster, more accurate delivery times at checkout directly increases customer conversion rates and reduces cart abandonment.

4. **Time-Based Trend Analysis: Revenue Contraction in 2018**
    - **Observation:** The line chart tracking the sum of price by year and month reveals a clear, steady downward trend in revenue generation as 2018 progresses, dropping from approximately 10K to under 5K.

- **Business Decision-Making:** This is a critical warning sign for executive leadership. Decision-makers must immediately investigate the root cause whether it is increased market competition, seasonal drop-offs, or a decline in website traffic. The business should launch targeted customer re-engagement campaigns, loyalty discounts, or new product lines to reverse this contraction.

5. **Future Recommendations: Freight Cost Optimization**
    - **Observation:** The dashboard highlights a total freight cost of 2.20M against a total revenue of 13.22M. Furthermore, freight values fluctuate based on the day of the week and product category.

    - **Business Decision-Making:** Shipping eats heavily into profit margins. The business should analyze which days incur the highest freight costs and encourage customers to select off-peak delivery windows (e.g., offering a small discount for "weekend delivery"). Additionally, for categories with disproportionately high freight costs, the business must evaluate whether to increase the product price to absorb the shipping cost or switch to a more cost-effective logistics provider.

AI Usage Acknowledgment

- In accordance with the assignment guidelines and the CLEAR framework, I acknowledge the use of generative AI (Google Gemini) as a supporting tool for learning and development during this project. Specifically, AI was utilized to:

    - Assist in debugging and correcting data aggregations within Power BI (e.g., resolving summing errors for delivery delays).

    - Refine and improve the English grammar and professional tone of the written explanations for the dashboard components.

    - Help brainstorm the structure for the business insights and recommendations.

- I confirm that the end-to-end technical implementation including data preparation, dimensional modeling, ETL pipeline creation, and dashboard design is entirely my own work.